

Question ID: 9ed9f54d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which of the following is equivalent to $2(x^2 - x) + 3(x^2 - x)$?

Answer

- A. $5x^2 - 5x$
- B. $5x^2 + 5x$
- C. $5x$
- D. $5x^2$

Correct Answer: A

Rationale

Choice A is correct. Since $(x^2 - x)$ is a common term in the original expression, like terms can be added:

$2(x^2 - x) + 3(x^2 - x) = 5(x^2 - x)$. Distributing the constant term 5 yields $5x^2 - 5x$.

Choice B is incorrect and may result from not distributing the negative signs in the expressions within the parentheses. Choice C is incorrect and may result from not distributing the negative signs in the expressions within the parentheses and from incorrectly eliminating the x^2 -term.

Choice D is incorrect and may result from incorrectly eliminating the x-term.